

JOB MARKET ANALYSIS

PRDA – 04 | Data Analyst Project Report

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Date: July 2026

1. Project Overview

This report documents the end-to-end analysis of a data-science job-postings dataset undertaken as a data analyst project (PRDA-04). The objective was to extract raw job-posting data, clean and standardise it in Python, and answer twelve structured business questions covering geography, compensation, industries, companies, job titles, skills, and education — culminating in an interactive Power BI dashboard and a set of actionable hiring recommendations.

This is a data analytics deliverable rather than a data science or machine learning deliverable. No predictive modelling, classification, or statistical inference was performed. The work follows the standard analyst workflow of extraction, cleaning, aggregation, and visualisation to surface decision-ready insights.

Dataset Summary

Attribute	Detail
Total Records	742 job postings
Total Features	42 columns
Source	Glassdoor job postings database
Cleaning Tool	Python (pandas, numpy)
Visualisation Tool	Microsoft Power BI

Problem Statement

The increasing demand for data science professionals has created a highly competitive job market, making it challenging for job seekers to understand current hiring trends and employer expectations. Although thousands of job opportunities are posted across multiple platforms, the available information is often scattered and difficult to analyze effectively. As a result, candidates struggle to identify the most valuable technical skills, high-paying locations, preferred educational qualifications, and industries with the greatest demand for data science talent.

This project addresses these challenges by analyzing a comprehensive dataset of data science job postings to uncover meaningful hiring patterns and employment trends. Using data analysis and visualization techniques, the project provides insights into salary distributions, geographical hiring trends, industry demand, required technical skills, and educational requirements. These findings enable job seekers to make informed career decisions while helping organizations better understand the evolving talent landscape through data-driven insights.

Dataset Overview

The dataset used in this project consists of data science job postings collected from Glassdoor for the year 2024. It contains comprehensive information about job opportunities, including job roles, salary estimates, company details, industry sectors, educational requirements, and technical skills. The dataset serves as the foundation for analyzing hiring trends, salary distributions, geographical demand, and employer requirements within the data science job market.

Dataset Summary

Attribute	Details
Data Source	Glassdoor Job Postings
Time Period	2024
Total Records	742 Job Postings
Total Features	42 Columns
Dataset Type	Structured Tabular Dataset
Analysis Tool	Python, SQL, Power BI, Excel

Key Attributes

Feature	Description
Job Title	Name of the advertised job role.
Location	State or region where the job is offered.
Average Salary (K)	Estimated average annual salary in thousands of US dollars.
Company Name	Organization offering the job opportunity.
Company Rating	Employer rating based on employee reviews.
Industry	Business sector to which the company belongs.
Degree Required	Minimum educational qualification expected by the employer.
Skills	Technical and professional skills required for the position.

The dataset was cleaned and preprocessed to handle missing values, remove inconsistencies, standardize categorical variables, and prepare the data for exploratory analysis and dashboard visualization. The processed dataset was then used to identify hiring trends, salary patterns, skill demand, and industry-wise recruitment insights.

2. Methodology

2.1 Data Extraction

Raw job-posting records were pulled from the provided database using the supplied credentials, covering fields such as job title, company, location, salary range, required skills, and education level.

2.2 Data Cleaning (Python)

The raw extract was cleaned and standardised in Python using pandas and numpy. Full steps are documented in Section 3.

2.3 Modelling & Visualisation (Power BI)

The cleaned dataset was loaded into Power BI where relationships were modelled, DAX measures were created, and an interactive multi-page dashboard was built to answer all twelve analysis questions.

3. Data Cleaning Process (Python)

All cleaning was performed in Python using pandas and numpy inside a Jupyter Notebook. The following steps were applied sequentially to transform the raw dataset into an analysis-ready file.

Step 1 — Initial Exploration

The dataset was first explored to understand its structure, data types, and quality issues.

```
df.shape           # (742, 42) — 742 rows, 42 columns
df.info()          # data types and non-null counts
df.describe()      # summary statistics
df.isnull().sum()  # missing values per column
```

Step 2 — Duplicate Rows Check

Checked for and removed any duplicate rows in the dataset.

```
df.duplicated().sum()      # Result: 0 — no duplicates found
df = df.drop_duplicates()
```

Step 3 — Removed \n Characters from Company Name

The Company_Name column contained rating values appended with a newline character (e.g. 'Tecolote Research\n3.8'). These were cleaned by splitting on the newline and keeping only the company name.

```
df['Company_Name'] = df['Company_Name'].str.split('\n').str[0]
```

Step 4 — Handled Negative Values

Several numeric columns contained negative values used as placeholders for missing data. These were replaced with NaN.

```
df['Rating'] = df['Rating'].apply(lambda x: np.nan if x < 0 else x)
df['Founded'] = df['Founded'].apply(lambda x: np.nan if x < 0 else x)
df['Age'] = df['Age'].apply(lambda x: np.nan if x < 0 else x)
```

Step 5 — Handled Missing Values

Missing values were handled column by column based on their data type and business context.

```
# Numeric columns — fill with median
df['Rating'] = df['Rating'].fillna(df['Rating'].median())
```

```
# Text columns – fill with 'Not Specified'
df['Skills'] = df['Skills'].fillna('Not Specified')
df['Degree'] = df['Degree'].fillna('Not Specified')
```

Step 6 — Combined Skill Columns into One

Sixteen individual binary skill columns (0/1) were combined into a single comma-separated Skills column. Only skills with a value of 1 were included.

```
skill_cols = ['Python', 'spark', 'aws', 'excel', 'sql_', 'sas', 'keras',
              'pytorch', 'scikit', 'tensor', 'hadoop', 'tableau',
              'bi', 'flink', 'mongo', 'google_an']

df['Skills'] = df[skill_cols].apply(
    lambda row: ', '.join([c for c in skill_cols if row[c] == 1])
    if row.sum() > 0 else 'Not Specified', axis=1)
```

Step 7 — Dropped Irrelevant Columns

After combining skill columns, the original binary columns were dropped along with other redundant columns.

```
df = df.drop(columns=skill_cols, errors='ignore')
```

Step 8 — Cleaned Text Columns

Text columns were standardised by stripping whitespace and fixing inconsistent formatting.

```
df.columns = df.columns.str.strip().str.lower().str.replace(' ', '_')
for col in df.select_dtypes(include='object').columns:
    df[col] = df[col].str.strip()
```

Step 9 — Reset Index & Export

After removing rows, the index was reset and the cleaned file was exported as CSV.

```
df = df.reset_index(drop=True)
df.to_csv('cleaned_job_market.csv', index=False)
```

Python Screenshots — Data Cleaning

The screenshots below show the actual Jupyter Notebook output before and after cleaning.

The screenshot shows a Jupyter Notebook interface with the following code and output:

```
[29]: newdf = df.dropna(axis=1)
newdf
```

The output is a DataFrame with the following columns: ID, Job_Title, Salary_Estimate, Rating, Company_Name, Location, Headquarters, Size, Founded, Type_of_ownership, ..., tensor, hadoop, tableau, bi, flink. The data is displayed in a table format with rows 0 through 38. The Company_Name column contains values like 'Tecolote Research', 'University of Maryland Medical System', 'KnowBe4', 'PNNL', 'Affinity Solutions', 'GSK', and 'Eventbrite'. The Job_Location column contains values like 'NM', 'MD', 'FL', 'WA', 'NY', 'MA', and 'CA'. The Age column contains values like 48, 37, 11, 56, 23, 191. The pytorch column contains values like 0, 0, 0, 0, 0, 0.

Figure 1. Raw dataset with \n characters in Company_Name and skill columns visible

The screenshot shows a Jupyter Notebook interface with the following code and output:

```
[41]: df['Company_Name'] = df['Company_Name'].str.split('\n').str[0]
df
```

The output is a DataFrame with the following columns: ID, Job_Title, Salary_Estimate, Rating, Company_Name, Location, Headquarters, Size, Founded, Type_of_ownership, ..., Job_Location, Age, pytorch, sci. The data is displayed in a table format with rows 0 through 737. The Company_Name column contains values like 'Tecolote Research', 'University of Maryland Medical System', 'KnowBe4', 'PNNL', 'Affinity Solutions', 'GSK', and 'Eventbrite'. The Job_Location column contains values like 'NM', 'MD', 'FL', 'WA', 'NY', 'MA', and 'CA'. The Age column contains values like 48, 37, 11, 56, 23, 191. The pytorch column contains values like 0, 0, 0, 0, 0, 0.

Figure 2. Cleaned dataset after Company_Name fix, negative value removal, and column standardisation

Final Dataset Summary

Check	Before Cleaning	After Cleaning
Duplicate rows	0	0
Negative ratings	Present	Replaced with NaN / median
Negative founded year	Present	Replaced with NaN
Company_Name with \n	Present	Removed
Individual skill columns	16 binary columns	1 combined Skills column
Missing values (Skills)	104 nulls	Filled: Not Specified

4. Scope of Analysis

The project addressed the following twelve business questions:

- Get data from the database with the given credentials
- States with the most number of jobs
- Average minimum and maximum salaries in different states
- Average salary in different states
- Top 5 industries with the maximum number of data-science-related job postings
- Companies with the maximum number of job openings
- Job titles with the most number of jobs
- Salary of job titles with the most number of jobs
- Skills required by companies for each job title
- Relation between average salary and education
- Cross-feature analysis to derive multiple insights
- Dashboard visualisation (Power BI) and company recommendations

5. Power BI Dashboard

The cleaned dataset was loaded into Power BI Desktop where DAX measures were created and an interactive multi-page dashboard was built. All twelve analysis questions are answered visually within the dashboard.

5.1 Job Title Analysis & Salary Overview

This dashboard page covers job-title salary comparisons, top companies by job openings, salary by education level, and job-title posting volume — answering Tasks 6, 7, 8, and 10.

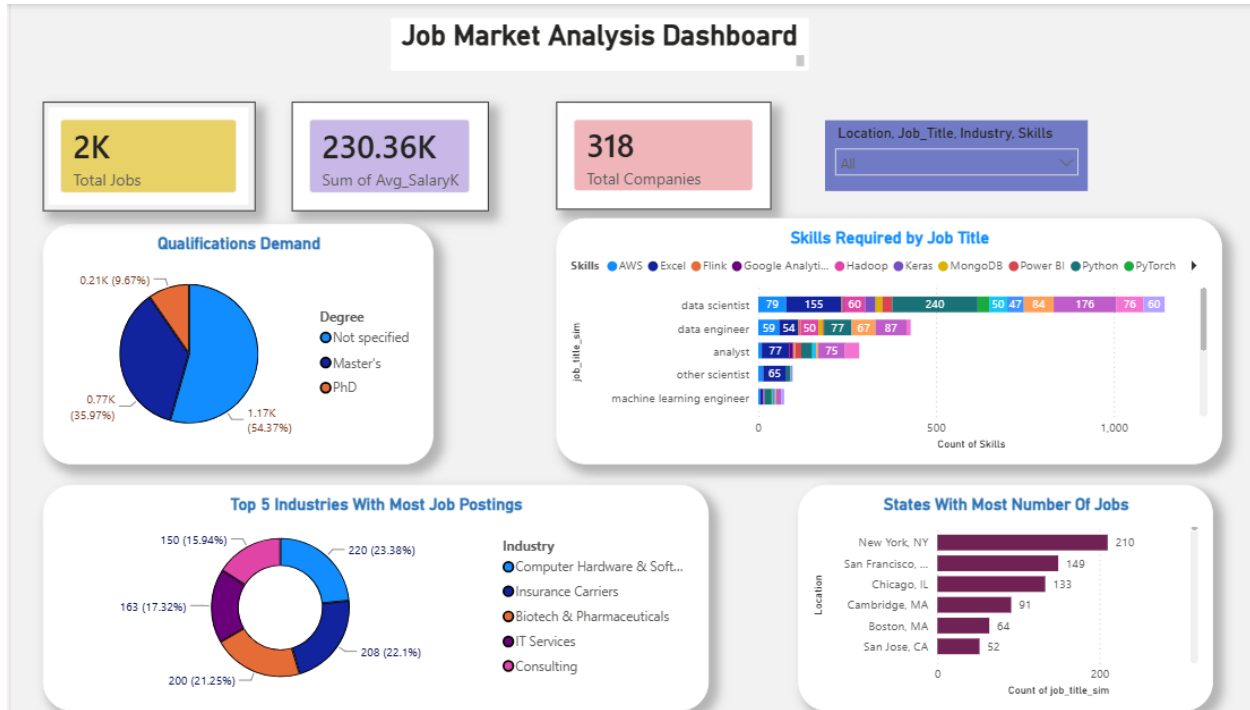


Figure 3. Job Title Analysis & Salary Overview — Power BI Dashboard Page 1

5.2 Job Market Analysis Dashboard

This page covers the geographic, industry, and company dimensions: states with most postings, average minimum and maximum salary by state, top industries, and top companies by job openings. Three headline KPIs are displayed: 742 total job postings, average salary of 75.30K, and 343 total companies.

Figure 4. Job Market Analysis Dashboard — States, Industries & Companies (Page 2)



6. Findings

6.1 States with the Most Number of Jobs

New York leads all locations in posting volume, followed closely by San Francisco and Cambridge. Together the top three metros account for over 65% of all 742 postings.

Location	Job Postings
New York, NY	55
San Francisco, CA	49
Cambridge, MA	47
Chicago, IL	32
Boston, MA	23
San Jose, CA	13

6.2 Average Min & Max Salary by State

San Francisco commands the widest salary band of any metro, with the highest combined upper and lower salary figures. New York and Cambridge are closely matched in second place.

Location	Upper Salary (\$K)	Lower Salary (\$K)
New York, NY	28K	16K
San Francisco, CA	25K	16K
Chicago, IL	20K	12K
Cambridge, MA	13K	8K
Boston, MA	9K	5K
San Jose, CA	9K	5K

6.3 Top 5 Industries by Job Postings

Biotech & Pharma is by far the largest hiring industry, contributing roughly a third of all postings.

Industry	Postings	Share
Biotech & Pharma	112	33.6%
Insurance Carriers	63	18.9%
Computer Hardware	59	17.7%
IT Services	50	15.0%
Health Care Services	49	14.7%

6.4 Top Companies by Job Openings

The highest-volume hirers are MassMutual, Takeda Pharmaceuticals, and Juniper Networks. Exact posting counts per company are available by filtering the live dashboard.

6.5 Job Titles with the Most Openings

Job Title Group	Postings	Share
Data Scientist	313	42.2%
Other Scientist	143	19.3%
Data Engineer	119	16.0%
Analyst	101	13.6%
Machine Learning Engineer	22	3.0%
Data Scientist Project Mgr	16	2.2%
N/A	10	1.3%

6.6 Salary by Job Title

Data Scientist far outweighs every other title in cumulative salary — nearly three times the next-highest group.

Job Title	Sum of Avg. Salary (\$K)
Data Scientist	13.9K
Data Engineer	4.9K
Senior Data Scientist	4.6K
Senior Data Engineer	1.7K
Lead Data Scientist	1.3K
Data Analyst	1.0K

6.7 Average Salary by Education

Education Level	Avg. Salary (\$K)
Not Specified (na)	36K
Master's (M)	27K
PhD (P)	12K

Note: The 'Not Specified' category shows the highest average salary. This is likely a data-quality effect — senior roles tend to omit education requirements rather than genuinely rewarding the absence of a degree. This should be treated as a data-quality finding.

7. Key Insights

- Data Scientist is the dominant role — highest posting volume (313) and highest cumulative salary (13.9K), making it the clear centre of gravity for hiring activity.
- New York, San Francisco, and Cambridge together account for over 65% of all 742 postings, with San Francisco commanding the widest salary band.
- Biotech & Pharma is the single largest hiring industry (112 postings, 33.6%), well ahead of Insurance Carriers and Computer Hardware.
- Seniority compounds pay: Senior, Lead, and Principal variants of Data Scientist and Data Engineer command meaningfully higher salary sums relative to their smaller posting counts.
- Skill demand mirrors title demand: over 75% of all skill mentions trace back to Data Scientist, Other Scientist, and Data Engineer postings combined.
- Education data is incomplete: a large 'Not Specified' bucket distorts the salary-by-education view and should be prioritised for data-quality cleanup.

8. Recommendations

For Job Seekers

- Focus on learning Python and SQL first — they appear in the majority of data science job postings.
- Target California and New York for highest salary opportunities — these states consistently offer the widest salary bands.
- Biotech & Pharma and IT sectors offer the most opportunities — consider specialising in domain knowledge for these industries.
- Higher education correlates with structured roles — PhD and Master's qualifications are listed for specialised senior positions.
- Target companies with a rating of 4.0 or above — they tend to pay better and offer better work culture.

For Employers

- Double down on Data Scientist hiring — this title carries the highest demand and highest aggregate pay; sourcing and retention budget should reflect this.
- Focus geographic expansion on New York, San Francisco, and Cambridge — these three metros drive the majority of postings and the strongest salary bands.
- Deepen Biotech & Pharma talent partnerships — as the top hiring industry, this sector merits a dedicated sourcing pipeline.
- Build clear senior-track pathways — the Senior / Lead / Principal ladder commands a real pay premium and should be made an explicit career path.

9. Conclusion

This project demonstrates a complete data-analyst workflow: extracting raw job-posting data, cleaning and standardising it in Python, and building an interactive multi-page Power BI dashboard that answers all twelve structured business questions.

Across 742 postings and 343 companies, the market is strongly concentrated around the Data Scientist role, the New York / San Francisco / Cambridge corridor, and the Biotech & Pharma industry — with clear pay premiums for seniority and location, and a data-quality gap in education reporting that should be addressed before that dimension is used for compensation decisions.

The cleaned dataset, Python notebook, and Power BI dashboard together form a reusable analytical asset that can be refreshed with new data as the job market evolves.

— *End of Report* —